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Article

Low-Level Laser Mitigates Cisplatin-Induced Oxidative Stress and Apoptosis via AMPK-Mediated Autophagy in PC12 Cells

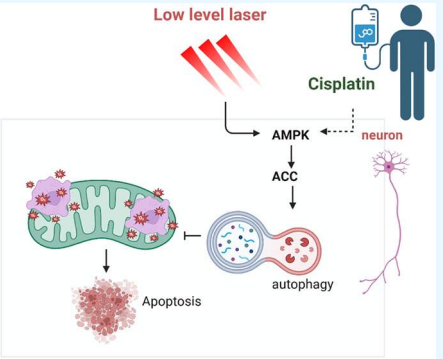
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ABSTRACT: Platinum-based chemotherapeutic agents, such as cisplatin, form the cornerstone of many modern anticancer treatment regimens. Despite their clinical effectiveness, long-term use is often limited by dose-related side effects, notably peripheral neuropathy, which significantly impacts patient quality of life and restricts treatment success. Neuroprotective strategies are still limited. Low-level laser (LLL) therapy, also known as photobiomodulation, has recently gained recognition as a noninvasive approach capable of reducing various tissue damages, including nerve injuries. However, the cellular and molecular mechanisms behind its protective effects in chemotherapy-induced neurotoxicity are not yet fully understood. In this study, we examined the neuroprotective effects of LLL in an in vitro model of cisplatin toxicity using PC12 neuronal cells. PC12 neuronal cells were pretreated with LLL irradiation at 660 nm and 5 J/cm^2 for 10 min prior to exposure to 10 μM cisplatin for 24 h. Cell viability and cytotoxicity were assessed using MTT and LDH assays. Apoptosis was evaluated via TUNEL staining and caspase-3 activity. Mitochondrial function and oxidative stress were examined by JC-1 staining for membrane potential, MitoSOX for mitochondrial ROS, and MDA assay for lipid peroxidation. AMPK expression was silenced using siRNA, and protein levels related to autophagy, apoptosis, and AMPK signaling were analyzed by Western blotting. Our results showed that pretreatment with LLL at 5 J/cm^2 significantly improved cell viability and decreased cisplatin-induced cytotoxicity and apoptosis at a concentration of 10 μM. LLL reduced oxidative stress caused by cisplatin and maintained mitochondrial function, as demonstrated by lower lipid peroxidation, decreased mitochondrial ROS production, and stabilization of mitochondrial membrane potential. In addition, we found that these protective effects were mediated through activation of the AMPK signaling pathway, which enhanced autophagy and inhibited apoptotic processes. Silencing of AMPK eliminated the neuroprotective effects of LLL, confirming its key role in this mechanism. Overall, our findings suggest that LLL therapy exhibits robust neuroprotective effects against cisplatin-induced neuronal damage through AMPK-driven autophagy, offering a promising adjunctive strategy for managing chemotherapy-induced peripheral neuropathy.



1. INTRODUCTION

Chemotherapy-associated toxicities have emerged as a significant area of research, driven by the growing population of cancer survivors resulting from advances in cancer therapy. Cisplatin is a platinum-based chemotherapy drug extensively used, either alone or in combination with other drugs, for treating a broad spectrum of malignancies. Nevertheless, its administration is associated with a range of adverse effects, including peripheral neurotoxicity[1] and cognitive deficits (also known as chemo-brain).[2] Cisplatin-induced peripheral neurotoxicity presents with sensory disturbances such as numbness, tingling, burning, altered sensitivity, and pain, commonly affecting the hands and feet in a "glove-and-stocking" pattern.[1] These symptoms may persist for several months and can progressively deteriorate even after the cessation of treatment, a phenomenon referred to as "coasting."[3] On the other hand, neuroimaging studies have identified structural and functional brain changes associated with chemotherapy in patients experiencing cognitive impairments such as difficulty multitasking, memory lapses, and confusion.[4] While the precise mechanism remains unclear, oxidative stress,[5] mitochondrial dysfunction,[6,7] and apoptosis[8] have been suggested to be implicated in cisplatin-induced neurotoxicity. As chemotherapy-induced neuropathy represents a leading cause of chemotherapy dose reduction and treatment discontinuation, exploring effective strategies to mitigate the neurotoxic effects of cisplatin has become a critical area of investigation.

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Low-level laser (LLL) irradiation, also known as photobiomodulation therapy, has been widely recognized in clinical medicine for over four decades as a noninvasive treatment used to promote wound healing and alleviate inflammation, edema, and pain.

Over the past few years, the utilization of LLL has emerged as a noteworthy subject of exploration within oncologic settings for the management of cancer treatment toxicities, such as oral mucositis, lymphedema, and radiodermatitis.[9] Additionally, several studies have demonstrated the beneficial effects of LLL irradiation on the alleviation of neuropathic pain and the acceleration of functional recovery following chronic constrictive injury or complete resection of the sciatic nerve.[10] LLL therapy has also been shown to improve oxaliplatin-induced neuropathic pain in animal models,[11] as well as sensory symptoms, balance, and gait speed in patients with chemotherapy-induced peripheral neuropathy.[12] Besides, a recent study demonstrated that LLL diminished cisplatin-induced decreases in cell viability of PC12 neuronal cells and reduced apoptosis triggered by cisplatin.[13] These studies suggest that LLL treatment does not exert additional detrimental effects on cancer patients and may play a supportive role in promoting neural repair and regeneration. Nonetheless, the neuroprotective properties of LLL require further investigation to elucidate the underlying mechanisms.

Recognized as the principal energy sensor, adenosine monophosphate-activated protein kinase (AMPK) monitors cellular nutrient status and ensures energy homeostasis.[14] Numerous studies have shown that AMPK is associated with mitochondrial function and pain progression.[15] In chemotherapy-induced peripheral neuropathy, the significance of mitochondrial dysfunction[16] and AMPK activation[17] has been established, and the upregulation of AMPK through concomitant metformin treatment during chemotherapy has been shown to prevent the onset of chemotherapy-induced peripheral neuropathy.[18] Furthermore, AMPK has been identified as a key regulator of autophagy[19] and apoptosis[20] in nerve cells, and a recent study demonstrated that activation of autophagy ameliorated paclitaxel-induced mitochondrial damage and peripheral neuropathy during chemotherapy.[21] In addition, a couple of investigations have reported that LLL irradiation prevents doxorubicin-induced skeletal muscle atrophy[22] or periodontitis infection[23] by modulating mitochondrial function, apoptosis, anti-inflammatory responses, and antioxidant defenses via the upregulation of AMPK. Also, LLL therapy has been shown to exert neuroprotective effects through the activation of AMPK, as it restores neuronal mitochondrial bioenergetics and attenuates apoptosis and mitochondrial oxidative stress following spinal cord injury[24] or in a mouse model of Alzheimer's disease.[25] Taken together, LLL appears to possess the potential to upregulate AMPK-mediated autophagy and modulate oxidative stress and apoptosis, but further validation is still warranted.

In the current study, we evaluated the effects of LLL on cisplatin-treated PC12 neuronal cells, followed by an assessment of AMPK signaling activation in response to cisplatin in both the absence and presence of LLL. Subsequently, we investigated how the activation of AMPK affects the neuroprotective capabilities of LLL, thereby laying the groundwork for future clinical use.

2. MATERIALS AND METHODS

2.1. Cell Culture and Reagents

Dulbecco's Modified Eagle Medium (DMEM), 5,58,6,68 tetraethylbenzimidazolcarbocyanine iodide (JC-1), 2",7"−dichlorofluorescein diacetate (DCF-DA), chloroquine, cisplatin, and MitoTEMPO were bought from Sigma (St. Louis, MO, USA). Fetal bovine serum (FBS), penicillin−streptomycin (PS), and L-glutamine were purchased from Thermo Scientific (Waltham, MA, USA). Anti-β-actin, anti-AMPK, anti-p-AMPK, anti-ACC, anti-p-ACC, anti-Beclin-1, and anti-LC-3I/II antibodies were all obtained from Santa Cruz Biotechnology (Santa Cruz, CA, USA). Anti-p62, anticytochrome C, anti-p53, HRP-conjugated antirabbit, and antimouse secondary antibodies were purchased from Transduction Laboratories (CA, USA). Rat pheochromocytoma PC12 cells were obtained from the American Type Culture Collection (ATCC) and cultured in DMEM supplemented with 10% heat-inactivated horse serum, 5% FBS, 1% PS, and 1% L-glutamine. Cells were maintained at 37 °C in a humidified atmosphere with 5% CO_2. The culture medium was renewed every 2 days.

2.2. LLL and Cisplatin Treatment

An AlGaInP diode laser (AM-800, Konftec Co., Taipei, Taiwan) was used in this study. The laser probe was positioned vertically 15 cm above the culture dish. Laser irradiation was applied at a wavelength of 660 nm for 1 h (5 J/cm²). After 1 h of LLL irradiation, cisplatin (10 μM) was then added to PC12 cells for 24 h of stimulation without LLL irradiation. The 10 μM cisplatin dose was chosen because it was near the IC50 in our cell model. Additionally, it produced consistent levels of cytotoxicity enough to cause cellular injury without excessive cell death, allowing for the assessment of the intervention's protective effects and underlying mechanisms.

2.3. MTT and Lactate Dehydrogenase (LDH) Assay

Cell viability was determined by the MTT [3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide] assay. PC12 cells (1×10^4 cells) were seeded in 24-well plates and treated with cisplatin for 24 h. In some wells, LLL was performed before cisplatin stimulation. After treatment, 100 μL of MTT solution (5 mg/mL) was added to each well and incubated for 1 h at 37 °C. Formazan crystals were solubilized with 100 μL of DMSO, and absorbance was measured at 570 nm using a microplate reader (BioTek Synergy LX Multi-Mode Reader). Cell cytotoxicity was quantified by studying LDH released into the culture medium using a CyQUANT LDH Cytotoxicity Assay Kit (Thermo Fisher Scientific). After treatment, the medium was collected, and LDH activity was examined according to the manufacturer's protocol. The absorbance was measured at 490 nm with a reference wavelength of 680 nm.

2.4. Transfection of siRNA

Small interfering RNA (siRNA) targeting AMPKα and a negative control siRNA were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA). PC12 cells were seeded in 6-well plates and transfected at 60−70% confluency using a commercial siRNA transfection reagent kit, according to the manufacturer's instructions. Knockdown efficiency was assessed 48 h post-transfection using Western blot analysis of AMPK protein expression.

2.5. Western Blot Analysis

The total cell lysates were extracted using RIPA buffer containing protease and phosphatase inhibitors. Equal amounts of protein were separated by SDS-PAGE and transferred to PVDF membranes. Blots were incubated with the primary antibodies. HRP-conjugated secondary antibodies were applied, and signals were detected using an ECL substrate. Densitometry was performed using ImageJ software and normalized to β-actin.

2.6. Investigation of Mitochondrial Membrane Potential

Mitochondrial membrane potential ($\Delta\psi$m) was assessed using the JC-1 assay. After stimulation with cisplatin for 24 h, cells were stained with 5 μM JC-1 at 37 °C for 30 min. JC-1 aggregates (red fluorescence, FL2) indicate polarized mitochondria, while monomers (green fluorescence,

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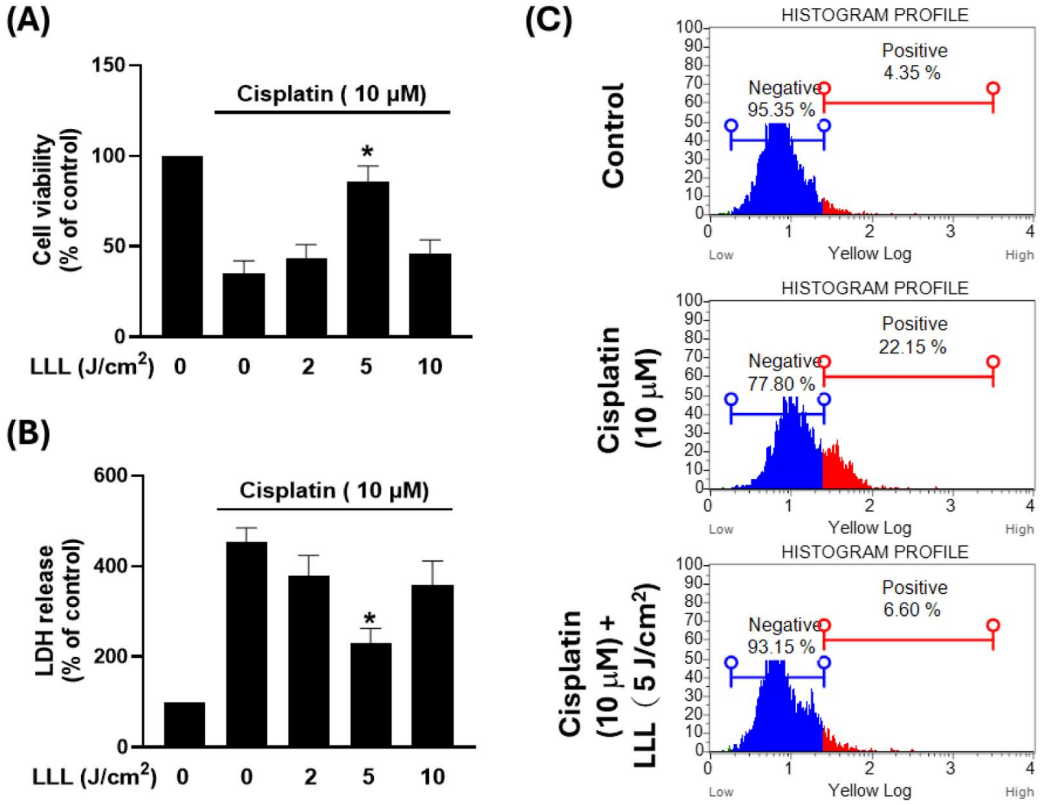


Figure 1. Low-level laser (LLL) exposure improves cisplatin-induced cytotoxicity and apoptosis in PC12 cells. Cisplatin-induced reduction in cell viability (A) and increase in cytotoxicity (B) were markedly attenuated by LLL treatment at 5 J/cm^2. (C) Apoptosis triggered by cisplatin was reduced by LLL irradiation, as determined by TUNEL assay with flow cytometry. Results are expressed as means ± SD of three independent experiments. * Indicates $p < 0.05$ compared with the control group.

FL1) represent depolarized mitochondria. Cells were analyzed by flow cytometry (BD FACSCalibur), and FL1/FL2 percentages were recorded.

2.7. Lipid Peroxidation and Oxidative Stress Measurement

Lipid peroxidation was quantified using a malondialdehyde (MDA) assay kit (Abcam) according to the manufacturer's protocol. Additionally, mitochondrial ROS was detected using MitoSOX Red (5 μM) for 30 min at 37 °C, followed by flow cytometry.

2.8. Apoptosis Analysis

DNA fragmentation was evaluated using the In Situ Cell Death Detection Kit, Fluorescein (Roche Diagnostics, Mannheim, Germany). After stimulation with cisplatin for 24 h, cells were fixed with 4% paraformaldehyde for 30 min at 37 °C and permeabilized with 0.1% Triton X-100, followed by incubation with the TUNEL reaction mixture for 1 h at 37 °C in a humidified chamber. Caspase-3 activity was measured using a Caspase-3 Assay Kit (BioVision, CA, USA). TUNEL-positive and Caspase-3-positive cells were quantified by flow cytometry.

2.9. Statistical Analysis

All experiments were independently repeated at least three times. Data are presented as the mean ± standard deviation (SD). Statistical comparisons between multiple groups were conducted using one-way analysis of variance (ANOVA), followed by Tukey's post-hoc test for pairwise group comparisons. A p-value of less than 0.05 was considered statistically significant.

3. RESULTS

3.1. LLL Irradiation Ameliorates Cisplatin-Induced Neurotoxicity

As shown in Figure 1A, LLL irradiation enhanced cell survival following cisplatin treatment, with an effective dose of 5 J/cm^2. Similarly, LLL treatment at an optimal dose of 5 J/cm^2 markedly reduced cisplatin-induced cytotoxicity, as determined by the LDH assay (Figure 1B). Using TUNEL flow cytometry, we found that the induction of apoptosis by cisplatin (10 μM) was dramatically attenuated by LLL irradiation (5 J/cm^2) (Figure 1C). As such, an efficacious dose of LLL (5 J/cm^2) was selected and used in the following experiments based on its ability to restore cell viability.

3.2. LLL Promotes Autophagy by Activating AMPK/ACC Signaling

In light of the fact that AMPK activation is critical in cisplatin-induced neurodamage[26] and is involved in the protective effects of LLL therapy against doxorubicin-induced muscle atrophy,[22] we sought to evaluate whether AMPK signaling participates in the neuroprotective properties of LLL treatment. First, we observed an increase in the level of AMPK phosphorylation in PC12 cells exposed to LLL irradiation. Moreover, LLL further potentiated the phosphorylation of AMPK induced by cisplatin in PC12 cells (Figure 2A,B). AMPK activation was also confirmed by the phosphorylation of acetyl-CoA carboxylase (ACC), which is a direct substrate of the AMPK stress-response pathway. Given the established role of the AMPK/ACC axis in regulating autophagy and mitochondrial function,[27] we thereafter evaluated the expression of phospho-ACC and autophagy-related markers. Our results showed that LLL irradiation amplified the cisplatin-induced phosphorylation of ACC, an effect that was abolished upon AMPK silencing (Figure 2C,D). Likewise, the cisplatin-induced increase in Beclin-1 and LC3-I/II levels and the decrease in p62 expression were enhanced by LLL treatment, and these effects were eliminated when AMPK

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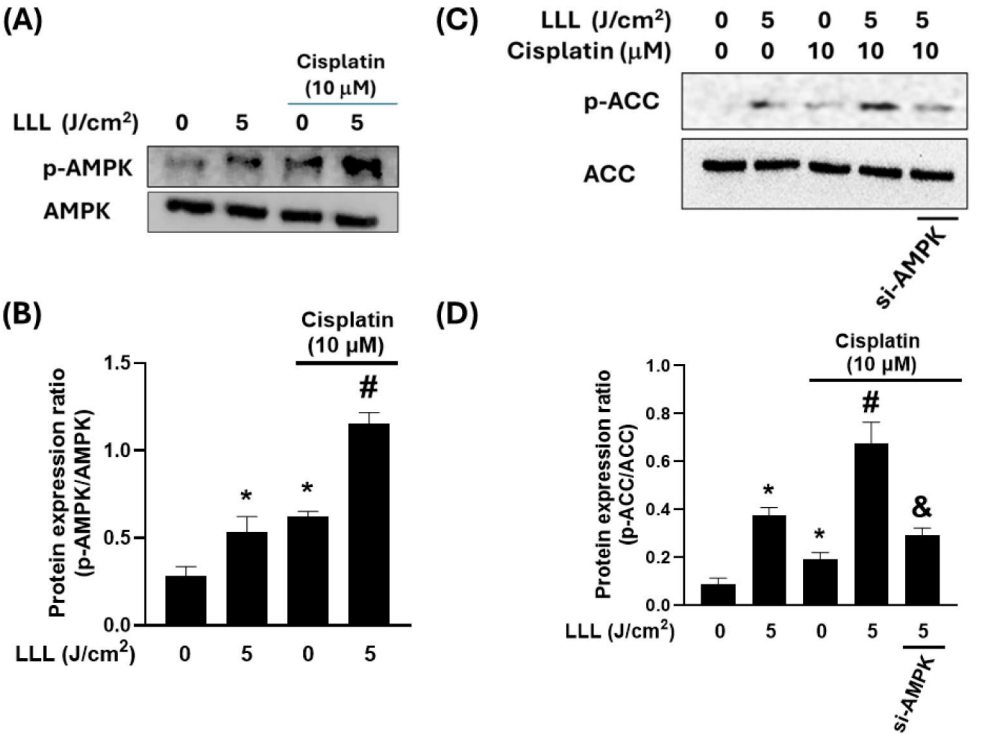


Figure 2. LLL potentiates the cisplatin-elicited activation of AMPK signaling. Representative Western blot images and corresponding densitometric bar graphs of phosphorylated AMPK (p-AMPK) and total AMPK in PC12 cells treated with 5 J/cm^2 LLL, either alone or in combination with cisplatin (A, B), are shown. Representative Western blot images and corresponding densitometric bar graphs of phosphorylated ACC (p-ACC) and total ACC in PC12 cells exposed to 5 J/cm^2 LLL, either alone, in combination with cisplatin, or under AMPK silencing conditions, are presented (C, D). Results are expressed as means ± SD of three independent experiments. * Indicates $p < 0.05$ compared with the control group. # Indicates $p < 0.05$ compared to the cisplatin group. & Indicates $p < 0.05$ compared to the cisplatin + LLL group.

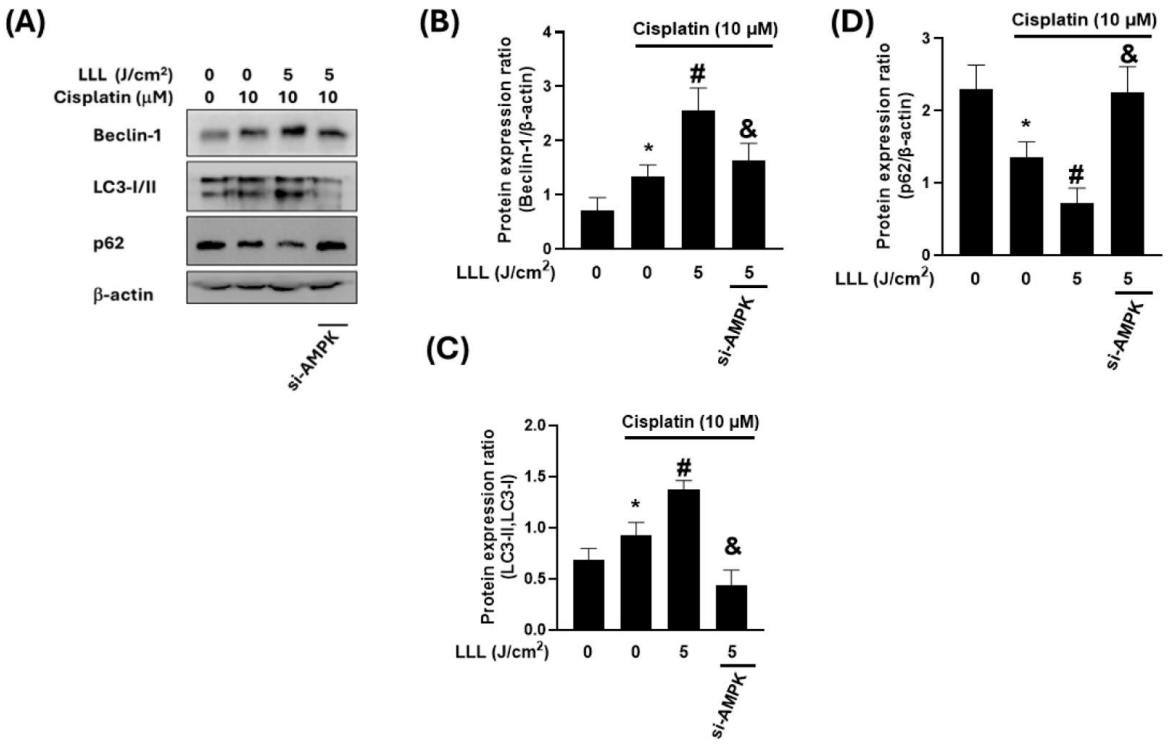


Figure 3. LLL enhances the cisplatin-induced autophagy in an AMPK-dependent manner. Representative Western blot images (A) and corresponding densitometric bar graphs (B, C, D) of Beclin-1, LC3-I/II, and p62 in PC12 cells exposed to cisplatin, either alone, in combination with 5 J/cm^2 LLL, or under AMPK silencing conditions, are presented. Results are expressed as means ± SD of three independent experiments. * Indicates $p < 0.05$ compared with the control group. # Indicates $p < 0.05$ compared to the cisplatin group. & Indicates $p < 0.05$ compared to the cisplatin + LLL group.

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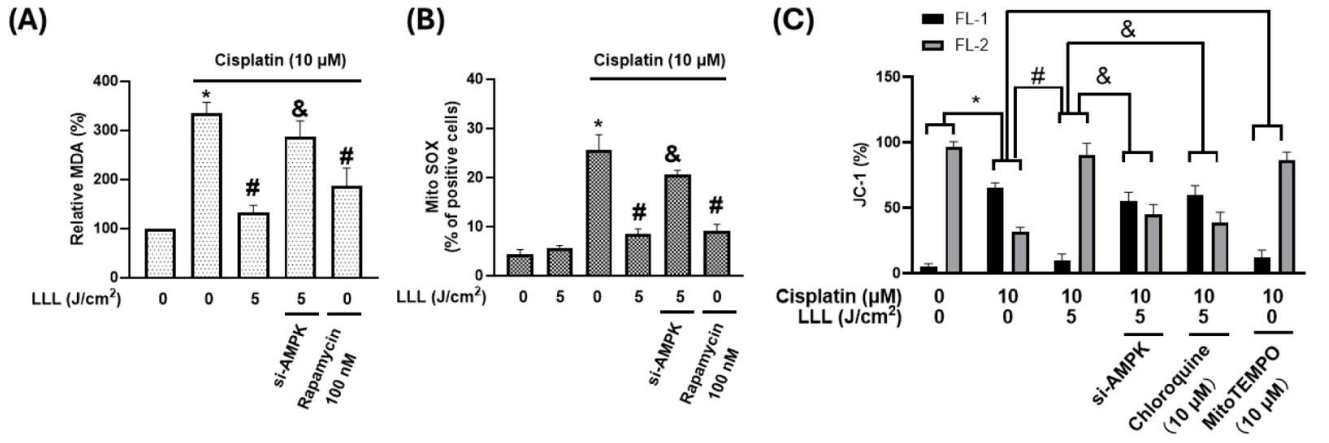


Figure 4. LLL reduces oxidative stress and changes in mitochondrial membrane potential evoked by cisplatin through AMPK-mediated autophagy. Representative measurements of (A) malondialdehyde (MDA) and (B) mitochondrial superoxide levels in PC12 cells treated with cisplatin, and further subjected to LLL, AMPK knockdown, or rapamycin-induced autophagy. Results were expressed as means ± SD of three independent experiments. * Indicates $p < 0.05$ compared with the control group. # Indicates $p < 0.05$ compared to the cisplatin group. & Indicates $p < 0.05$ compared to the cisplatin + LLL group. Mitochondrial membrane potential was assessed using JC-1 staining. Fluorescence intensities in the FL-1 (green) and FL-2 (red) channels were measured by flow cytometry and quantified, with the results presented as a bar graph (C). Experimental conditions included treatment with chloroquine (an autophagy inhibitor) and MitoTEMPO (a mitochondria-targeted antioxidant), either alone or in combination with cisplatin and/or LLL.

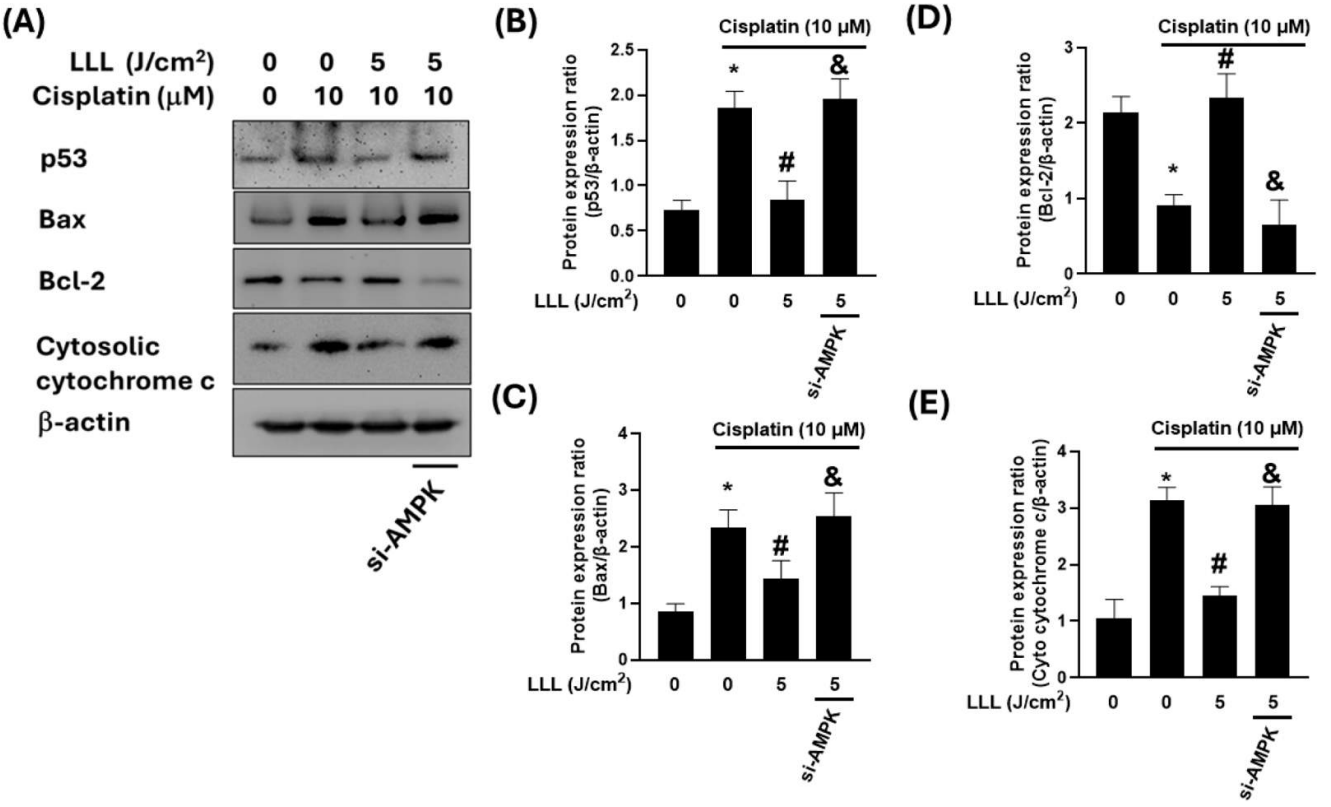


Figure 5. LLL inhibits cisplatin-induced apoptosis through the activation of AMPK. Representative Western blot images (A) and corresponding densitometric bar graphs (B-E) of p53, Bax, Bcl-2, and cytosolic cytochrome c in PC12 cells exposed to cisplatin, either alone, in combination with 5 J/cm² LLL, or under AMPK silencing conditions, were presented. Results are expressed as means ± SD of three independent experiments. * Indicates $p < 0.05$ compared with the control group. # Indicates $p < 0.05$ compared to the cisplatin group. & Indicates $p < 0.05$ compared to the cisplatin + LLL group.

was inhibited (Figure 3A–D). Collectively, these results suggest that LLL irradiation may further augment cisplatin-induced autophagy in PC12 cells, possibly through the AMPK-ACC signaling pathway.

3.3. LLL Protects Against Cisplatin-Elicited Oxidative Stress and Mitochondrial Impairment by Modulating AMPK-Mediated Autophagy

Subsequently, we investigated whether the activation of autophagy by LLL irradiation could mitigate cisplatin-induced

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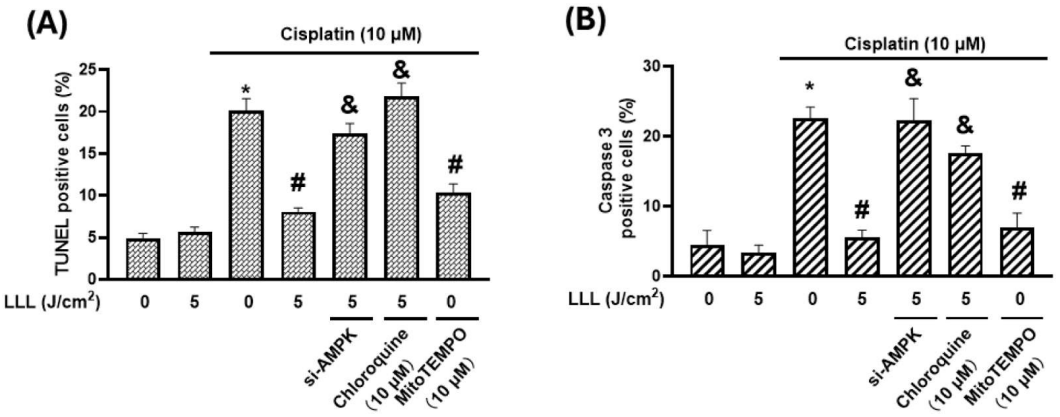


Figure 6. The antiapoptotic effect of LLL on cisplatin-treated cells is mediated through AMPK-dependent autophagy. Representative measurements of (A) TUNEL-positive and (B) caspase-3-positive cells treated with cisplatin, and further subjected to LLL, AMPK knockdown, chloroquine, or MitoTEMPO. Results were expressed as means ± SD of three independent experiments. * Indicates $p < 0.05$ compared with the control group. # Indicates $p < 0.05$ compared to the cisplatin group. & Indicates $p < 0.05$ compared to the cisplatin + LLL group.

oxidative stress and mitochondrial dysfunction through the AMPK pathway. Our results demonstrated that the cisplatin-elicited oxidative stress, measured by increased MDA levels, was blocked by LLL treatment (Figure 4A). These effects were abolished when AMPK was inhibited but reappeared upon rapamycin-induced autophagy, suggesting that AMPK-mediated autophagy may be involved in the mechanism by which LLL regulates oxidative stress. We also observed similar results when detecting mitochondrial reactive oxygen species (ROS) using MitoSOX mitochondrial superoxide indicators. The inhibitory effect of LLL on mitochondrial ROS disappeared upon AMPK silencing but was restored through rapamycin-induced autophagy (Figure 4B). Since cisplatin-induced mitochondrial dysfunction is related to oxidative stress and neurodamage,[28] we then measured the mitochondrial membrane potential by flow cytometry using JC-1 dye to verify whether the accumulation of oxidative stress contributes to mitochondrial impairment. Our data demonstrated that LLL irradiation in combination with cisplatin attenuated cisplatin-induced mitochondrial membrane potential depolarization; however, this effect was eliminated when AMPK was knocked down or when autophagy was suppressed by chloroquine treatment. As expected, treatment with MitoTEMPO, a mitochondrial superoxide scavenger that blocks mitochondrial and intracellular ROS generation, produced a result similar to that observed with LLL treatment (Figure 4C). These findings indicate that LLL confers protection against cisplatin-induced mitochondrial dysfunction by reducing the level of oxidative stress through AMPK-mediated autophagy.

3.4. LLL Attenuates Cisplatin-Induced Apoptosis by Regulating AMPK-Mediated Autophagy and Reducing Mitochondrial Oxidative Stress

Considering the central role of mitochondria in apoptosis, we sought to determine whether the neuroprotective effect of LLL against cisplatin-induced neurotoxicity is associated with its ability to mitigate mitochondrial dysfunction. We observed that cisplatin treatment led to an increase in the levels of p53, Bax, and cytosolic cytochrome c, accompanied by a decrease in Bcl-2 expression. While LLL treatment counteracted these changes, its effect was negated when AMPK was suppressed (Figure 5A–E). These findings demonstrate that AMPK activation is essential for the antiapoptotic effect of LLL in the context of cisplatin treatment.

We next investigated whether the inhibitory property of LLL on apoptosis involves the reduction of oxidative stress via AMPK-mediated autophagy. Assessment of TUNEL- and caspase-3-positive cells both indicated that apoptosis was triggered by cisplatin treatment. We observed that the apoptosis-reducing effect of LLL was undermined by AMPK deficiency or autophagy inhibition via chloroquine. Moreover, the administration of MitoTEMPO to block ROS generation resulted in outcomes comparable to those of LLL treatment alone (Figure 6A,B). Overall, the data demonstrate that the reduction of oxidative stress through AMPK-mediated autophagy is necessary for the suppressive effect of LLL irradiation on apoptosis during cisplatin treatment.

4. DISCUSSION

While the exact mechanisms underlying cisplatin-associated neuropathy remain to be fully elucidated, postmortem and experimental analyses have revealed that platinum is retained in the dorsal root ganglia, where it preferentially binds to DNA and is associated with reduced nuclear size.[29] Upon entering cells, cisplatin binds to mitochondrial DNA and forms adducts that cannot be repaired, leading to mitochondrial dysfunction.[6] Subsequently, the dysregulation of pathways involving ROS and apoptosis may lead to a gradual energy failure. Multiple studies have demonstrated that cisplatin treatment induces mitotoxicity and increases oxidative stress (e.g., elevated superoxide anion levels and lipid peroxidation), followed by apoptotic changes in the sciatic nerves, saphenous nerves, dorsal root ganglion, and spinal cord.[6,30] In the past few years, numerous randomized controlled trials have investigated the therapeutic efficacy of LLLs in individuals receiving neurotoxic chemotherapy. These studies have all demonstrated promising results in the prevention or reduction of chemotherapy-induced peripheral neuropathy in cancer patients undergoing chemotherapy[31] or in survivors experiencing neuropathic symptoms.[32] In the present study, we showed that LLL therapy may offer a viable management approach for established chemotherapy-induced neuropathy to prevent neuronal apoptosis, potentially via the modulation of AMPK-mediated autophagy.

Autophagy is a highly conserved intracellular degradation process that is essential for maintaining cellular homeostasis by removing damaged organelles, misfolded proteins, and other cytoplasmic components. Autophagy is tightly regulated by

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various signaling pathways, among which the AMPK and mammalian target of rapamycin (mTOR) pathways are central.[33] Cisplatin has been reported to trigger an autophagic response in various tumor cell lines as a protective mechanism against apoptosis, likely through AMPK signaling.[34] While it is known that cisplatin induces autophagy—acting either as a survival mechanism or as a pathway to cell death—its role appears to be context-dependent, with studies supporting both cytoprotective and cytotoxic functions.[35] As a matter of fact, a growing body of research has highlighted the pivotal role of AMPK-mediated autophagy in the pathogenesis of cisplatin-induced toxicity. For instance, multiple strategies aimed at enhancing AMPK-mediated autophagy have been demonstrated to alleviate cisplatin-induced acute kidney injury.[36] Consistently, AMPK-mediated autophagy induction has been reported to mitigate cisplatin-induced ototoxicity[37] and cognitive deficits.[38] In agreement with these studies, our results showed that cisplatin induces AMPK-mediated autophagy, and LLL treatment further augments autophagy to exert a protective effect and reduce mitochondrial dysfunction and oxidative stress, thereby ameliorating subsequent neuropathy.

It is well-known that the proper function of mitochondria is critical for maintaining several interrelated cellular pathways, such as ROS generation[39] and intracellular Ca^{2+} homeostasis.[40] The generation of ROS following cisplatin treatment is attributed to its direct impact on mitochondrial DNA, which disrupts the synthesis of proteins essential for the electron transport chain.[41] As such, cisplatin-induced mitochondrial DNA damage can lead to an elevated level of oxidative stress and calcium dysregulation, which may compromise the integrity of the mitochondrial outer membrane. This disruption facilitates the release of cytochrome *c* into the cytosol, thereby initiating caspase activation and promoting apoptotic cell death. Our results clearly demonstrate that cisplatin induces oxidative stress and mitochondrial dysfunction, as evidenced by increased levels of lipid peroxidation and mitochondrial superoxide indicators, along with alterations in mitochondrial membrane potential. Of note, these effects were diminished by LLL treatment, which was associated with reduced oxidative stress because of the activation of AMPK-mediated autophagy. Autophagosomes have been suggested to form membrane structures that encapsulate damaged or dysfunctional cellular components, deliver them to lysosomes for degradation, and contribute to mitochondrial turnover.[42] Consequently, autophagy is regarded as a critical cellular antioxidant pathway in neurodegenerative diseases, as the autophagic clearance of various damaged molecules may contribute to cellular antioxidant defenses.[43] Given the pivotal roles of oxidative stress and mitochondrial dysfunction in apoptosis, it is reasonable to propose that their attenuation could contribute to a reduction of apoptotic cell death. Indeed, several studies have suggested that the restoration of mitochondrial function and suppression of oxidative stress may help reduce apoptosis and mitigate cisplatin-induced neurotoxicity.[44] In line with these studies, our results indicated that LLL attenuated oxidative stress and mitochondrial impairment by modulating AMPK-mediated autophagy, ultimately leading to a reduced level of apoptosis.

Altogether, we showed that LLL treatment can be used as an adjunct therapy alongside chemotherapeutic agents to alleviate cisplatin-associated toxicity via the upregulation of AMPK-mediated autophagy. This study establishes a mechanistic foundation for the neuroprotective effects of LLL therapy in cisplatin-induced neuropathy and provides insights into its possible clinical benefits.

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Conceptualization: H.-C.O., K.-L.T., and P.-L.H.; methodology: H.-L.Y. and Y.-T.H.; software: H.-C.C. and W.-C.C.; validation: H.-C.O., K.-L.T., and P.-L.H.; formal analysis: H.-L.Y., Y.-T.H., and H.-C.C.; investigation: K.-L.T., H.-L.Y., Y.-T.H., and P.-L.H.; resources: H.-C.O.; data curation: H.-C.O. and K.-L.T.; writing: H.-C.O., K.-L.T., and P.-L.H. All authors have read and agreed to the published version of the manuscript. H.-C.O. and K.-L.T. equally contributed to this study.

Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

ACC	acetyl-CoA carboxylase
AMPK	adenosine monophosphate-activated protein kinase
APP/PS1	amyloid precursor protein/presenilin-1
ATCC	American Type Culture Collection
BDNF	brain-derived neurotrophic factor
CIPN	chemotherapy-induced peripheral neuropathy
CREB	cAMP response element-binding protein
DCF-DA	2′,7′-dichlorofluorescein diacetate
DMEM	Dulbecco's Modified Eagle Medium

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DMSO	dimethyl sulfoxide
DRG	dorsal root ganglion
eNOS	endothelial nitric oxide synthase
ETC	electron transport chain
FBS	fetal bovine serum
FOXO3a	forkhead box protein O3
HIF-1α	hypoxia-inducible factor 1-alpha
HMGB1	high-mobility group box 1
JC-1	5,5′,6,6′-tetrachloro-1,1′,3,3′-tetraethylbenzimidazolylcarbocyanine iodide
LC3-I/II	microtubule-associated protein 1*A*/1B-light chain 3
LDH	lactate dehydrogenase
LLL	low-level laser
MDA	malondialdehyde
MitoSOX	mitoSOX Red mitochondrial superoxide indicator
mTOR	mammalian target of rapamycin
MPTP	mitochondrial permeability transition pore
NF-κB	nuclear factor kappa B
PC12	rat pheochromocytoma-derived neuronal cell line
PGC-1α	peroxisome proliferator-activated receptor gamma coactivator 1-alpha
PS	penicillin−streptomycin
PVDF	polyvinylidene difluoride
ROS	reactive oxygen species
SD	standard deviation
SDS-PAGE	sodium dodecyl sulfate−polyacrylamide gel electrophoresis
siRNA	small interfering RNA
SIRT1	Sirtuin 1
TFAM	mitochondrial transcription factor A
TUNEL	terminal deoxynucleotidyl transferase dUTP nick-end labeling

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